

Algebra II/Trigonometry List of Final Exam Topics

2 hours, reference sheet provided, all scrap paper and reference sheet must be turned in with the exam.

Part A: 28 Questions, No Calculator allowed, 19 Multiple Choice, 9 Free Response

Part B: 11 Questions, Graphing Calculator allowed, 7 Multiple Choice, 4 Free Response

Chapter 6: Exponential and Logarithmic Functions

1. Graph exponential growth or decay functions and their transformations
2. Write models for situations involving exponential growth or decay
3. Compare regression models and determine the best fit for a data set
4. Solve exponential equations and inequalities
5. Convert between exponential and logarithmic form
6. Simplify and evaluate expressions using the properties of logarithms
7. Graph logarithmic functions and their transformations
8. Solve logarithmic equations and inequalities

Chapter 7: Rational Functions

9. Simplify rational expressions
10. Simplify complex fractions
11. Multiply and divide rational expressions
12. Determine the LCM of polynomials
13. Add and subtract rational expressions
14. Solve rational equations and inequalities
15. Graph rational functions with vertical and horizontal asymptotes

Chapter 8: Statistics and Probability

16. Use the binomial theorem to find the number of possible combinations for a specific binomial event
17. Use combinations and/or permutations to find the number of ways to achieve a particular outcome
18. Distinguish between sample surveys, experiments, and observational studies
19. Understand basic experimental design using random sampling, control group(s), experimental group(s), etc.
20. Make inferences about population parameters based on random samples of the population
21. Use the shapes of distributions to select appropriate statistics
22. Apply the empirical rule for a normal distribution

Chapter 9: Trigonometric Functions

23. Draw angles in standard position
24. Convert between degree measures and radian measures
25. Find coterminal angles and reference angles in degrees or radians
26. Understand the relationship between radian angle measures and the subtended arc length
27. Use a given trigonometric ratio or a given right triangle to find the values of other trigonometric ratios
28. Find values of trigonometric functions based on the unit circle
29. Use the properties of periodic functions to evaluate trigonometric functions
30. Graph trigonometric functions and their transformations including vertical translations, phase shifts, changes in amplitude, and changes to the period

Chapter 10: Trigonometric Identities and Equations

31. Use trigonometric identities to find trigonometric values
32. Use trigonometric identities to simplify expressions
33. Verify trigonometric identities
34. Solve trigonometric equations
35. Apply the Law of Sines and/or Law of Cosines to solve triangles and/or calculate the area of a triangle

Chapter 11: Sequences and Series

36. Distinguish between arithmetic and geometric sequences
37. Write the rule for the n th term of an arithmetic or geometric sequence
38. Convert a finite series, or partial sum, into sigma notation
39. Calculate the partial sum of an arithmetic or geometric series
40. Calculate the sum of an infinite geometric series, if possible

Review Materials

- Cumulative Review Tests (at the end of each chapter) - Pgs. 368 & 369, 462 & 463, 526 & 527, 588 & 589, 650 & 651, 690 & 691
- Extra Practice - Pgs. R6 to R10
- Old DeltaMath and textbook classwork/homework assignments
- Old DeltaMath practice quizzes/tests
- Old chapter quizzes and tests
- Final Exam Review DeltaMath assignment